EXPLAINABLE - DRIFT-AWARE - COST-ACCOUNTED

TabML-Guard Report — Electricity Price Movement

electricity · 45,312 rows × 7 features · classification

RECOMMENDED MODEL

logreg_l1

ROC_AUC (HELD-OUT)

0.7884

AT A GLANCE

- Simple model wins
- Tuning: marginal gain
- Inputs drifted, performance held

Headline finding

logreg_l1 is the recommended model, scoring 0.7884 roc_auc on a strictly held-out, time-ordered test set.

Recommendation

Hyperparameter tuning moved the score by less than 1% over defaults, so the default configuration is a defensible production choice. At an equal budget Bayesian search edged out random search, but cost 1.5x the wall-clock. The input distribution shifted materially (PSI above 0.25) while accuracy held up - monitor, but no retrain is urgent yet.

On model complexity

A regularized linear baseline matched or beat every ensemble here. That is a finding, not a failure: it says the signal in this dataset is largely additive, and it argues for shipping the cheaper, more interpretable model.

This report was produced automatically by the TabML-Guard pipeline. Every number below comes from a single run on a strictly held-out, time-ordered test split - no test data was used for model selection or tuning.

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STAGE 0

Dataset, profile and configuration

The pipeline begins by profiling the raw table and deciding, from the target column alone, whether this is a classification or a regression problem. When a time column is present the split is chronological - the model trains on the past and is scored on the future.

Rows	45,312
Features	7
Target column	class
Detected task	classification
Missing values	0
Time column	date
Time span	0.0 -> 1.0
Class balance	DOWN=0.576 UP=0.424

Configuration used

This is the entire dataset-specific input to the run. Everything else is generic pipeline code.

```
dataset.source        openml
dataset.name          electricity
dataset.path          -
dataset.target        class
dataset.time_col      date
dataset.drop_cols     []
split                 train=0.6  test=0.4
models.boosters       ['xgboost', 'lightgbm', 'catboost', 'random_forest']
models.stacking       True
tuning.budget_trials  40
tuning.cv_splits      3
drift.n_chunks        10
drift.retrain         sliding
random_state          42
```

STAGE 1

Regularized linear baselines

Before reaching for an ensemble, the pipeline establishes what a well-regularized linear model can do. This is the honesty check: if a linear model is competitive, the extra complexity of boosting has to justify itself.

model	roc_auc	accuracy	f1	precision	recall	fit
logreg_l1	**0.7884**	**0.7246**	**0.6221**	**0.7433**	**0.5349**	**0.2s**
logreg_elasticnet	0.7884	0.7246	0.6220	0.7431	0.5349	0.2s
logreg_l2	0.7884	0.7245	0.6220	0.7430	0.5349	0.0s
logreg_none	0.7884	0.7245	0.6220	0.7430	0.5349	0.0s

the 4 regularizers span only 0.0000 roc_auc (best 'logreg_l1' 0.7884) — with abundant rows and few features, penalties barely move the fit at the default strength; L1 path: at C=0.001 only 2/13 features survive (sparsity), rising to 13/13 as C grows

STAGE 2

Ensemble and boosting models

Gradient-boosted trees plus a stacking meta-model, all scored through the same evaluation path as the baselines so the leaderboard rows are directly comparable.

model	roc_auc	accuracy	f1	precision	recall	fit
logreg_l1	0.7884	0.7246	0.6221	0.7433	0.5349	0.2s
logreg_elasticnet	0.7884	0.7246	0.6220	0.7431	0.5349	0.2s
logreg_l2	0.7884	0.7245	0.6220	0.7430	0.5349	0.0s
logreg_none	0.7884	0.7245	0.6220	0.7430	0.5349	0.0s
catboost	**0.7814**	**0.7105**	**0.6621**	**0.6549**	**0.6695**	**2.0s**
lightgbm	0.7742	0.7007	0.6561	0.6394	0.6737	0.3s
xgboost	0.7739	0.7001	0.6579	0.6368	0.6803	0.3s
stacking	0.7730	0.7049	0.6544	0.6496	0.6592	21.9s
random_forest	0.7718	0.7085	0.6505	0.6612	0.6401	0.9s

Best ensemble: **catboost**. The strongest single booster (**catboost**) is the one carried into the tuning comparison in Stage 3.

STAGE 3

Hyperparameter optimization: benefit vs cost

Four strategies compete on **catboost**: library defaults, an exhaustive coarse grid, random search, and Bayesian optimization (Optuna TPE). Random and Bayesian are given an **equal budget** of 40 trials, which makes them a fair head-to-head. Crucially, the table reports not just the score but what each strategy *cost*.

strategy	test roc_auc	wall clock	model fits	cv score
default	0.7818	1.8s	1	-
grid	0.7844	12.2s	54	0.8480
random	0.7858	47.3s	120	0.8478
bayes	**0.7860**	**69.6s**	**120**	**0.8499**

The *cv score* column is the internal cross-validated score each search optimized (scikit-learn's convention is higher-is-better, so error metrics appear negated). The *test* column is the winning configuration refit once and scored on the untouched held-out period - that is the number to trust.

> **Is tuning worth it?**
>
> best tuner 'bayes' reached roc_auc=0.7860 vs default 0.7818 (+0.0043); at equal budget, Bayesian search won (roc_auc: bayes 0.7860 @ 69.6s vs random 0.7858 @ 47.29s)

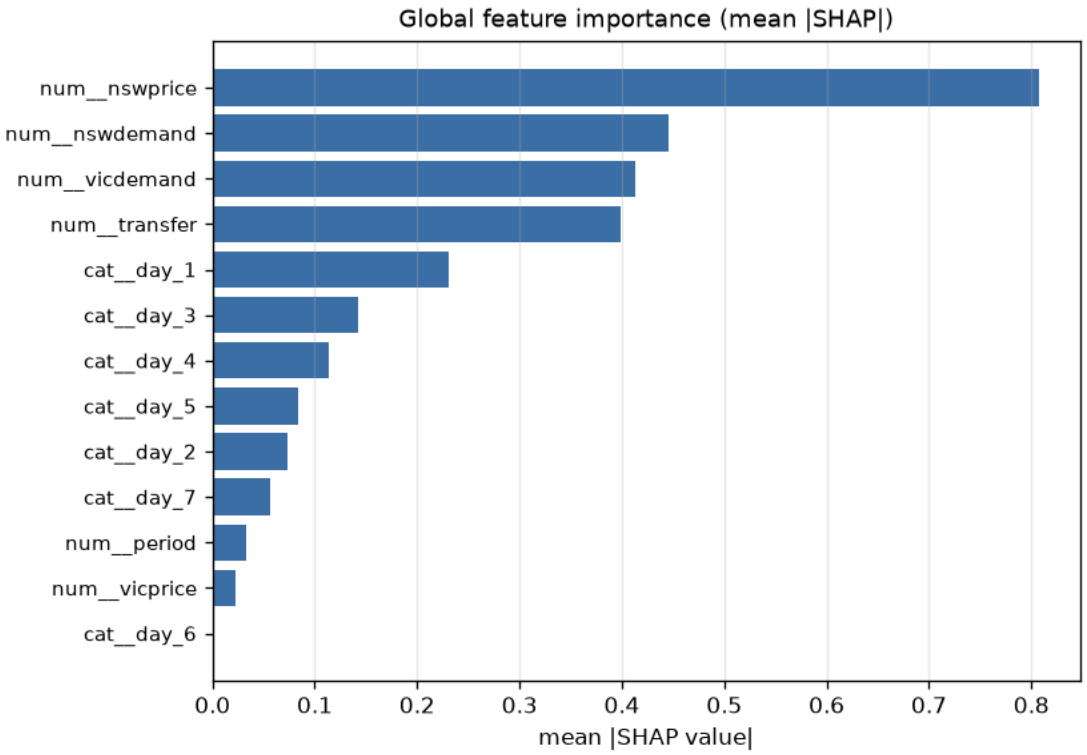
STAGE 4

Explainability and prediction reliability

SHAP attributions for **logreg_l1**, using the explainer matched to the model family (tree, linear, or model-agnostic fallback). Global importance answers *what drives this model overall*; the local example shows how a single prediction

was reached.

feature	mean |SHAP|
num__nswprice	0.80704
num__nswdemand	0.44571
num__vicdemand	0.41284
num__transfer	0.39894
cat__day_1	0.23082
cat__day_3	0.14209
cat__day_4	0.11411
cat__day_5	0.08416
cat__day_2	0.07290
cat__day_7	0.05672
num__period	0.03359
num__vicprice	0.02251
cat__day_6	0.00084



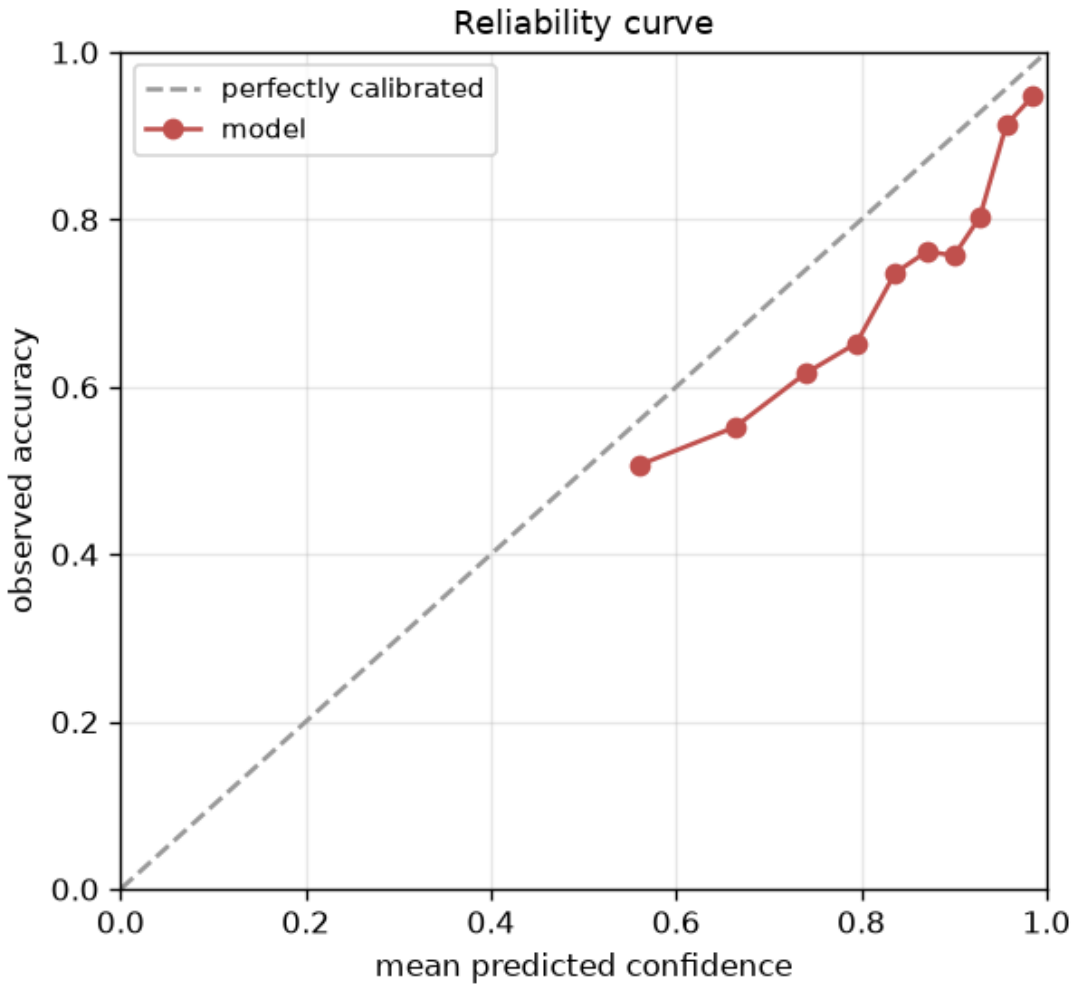
Global feature importance (mean |SHAP|)

Confidence-based triage

A model that knows when it is unsure is far more useful in production than one that is uniformly confident. Binning the held-out predictions by confidence shows whether confidence actually tracks correctness - and if it does, the least-confident slice can be routed to a human.

confidence band	n	mean confidence	accuracy
least confident 1/10	1,813	0.561	50.7%
2/10	1,812	0.663	55.2%
3/10	1,813	0.740	61.6%
4/10	1,812	0.795	65.2%
5/10	1,812	0.837	73.6%
6/10	1,813	0.871	76.2%
7/10	1,812	0.900	75.7%
8/10	1,813	0.928	80.3%
9/10	1,812	0.957	91.3%
most confident 10/10	1,813	0.985	94.8%

Flagged for review	20.0%
Of all errors, caught	34.2%
Accuracy on auto-approved	77.3%
Accuracy on flagged	52.9%
Confidence cut-off	0.705



Reliability curve: confidence vs observed accuracy

STAGE 5

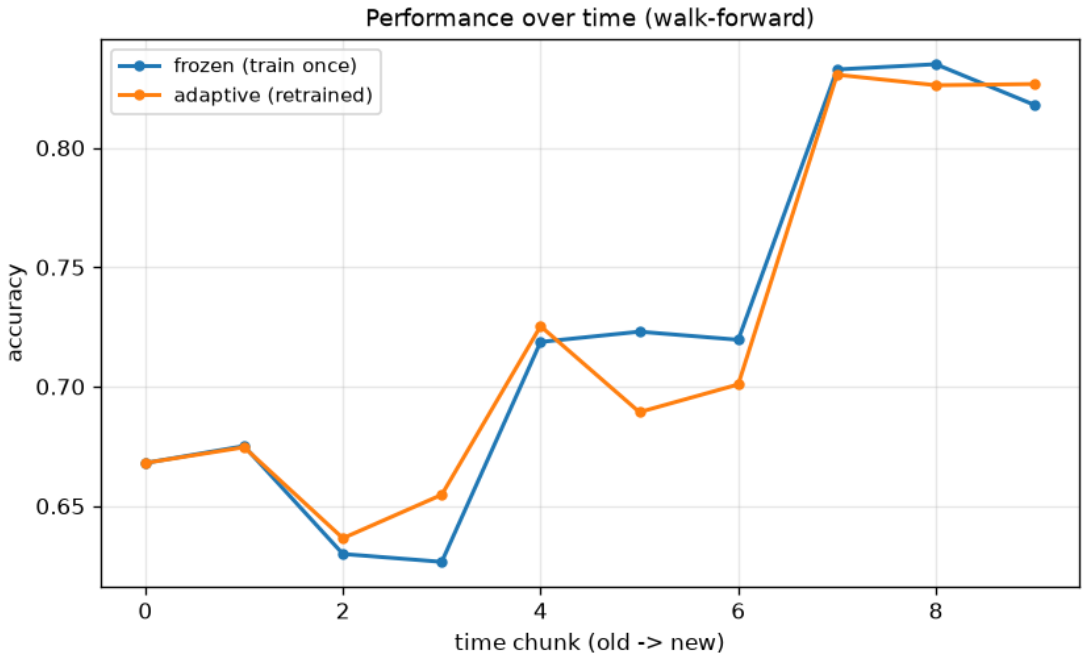
Concept drift and statistical significance

The held-out period is replayed in 10 chronological chunks. A **frozen** model (trained once, never updated) is compared against an **adaptive** model that predicts each chunk *before* learning from it - so there is no look-ahead. Retraining mode: **sliding**.

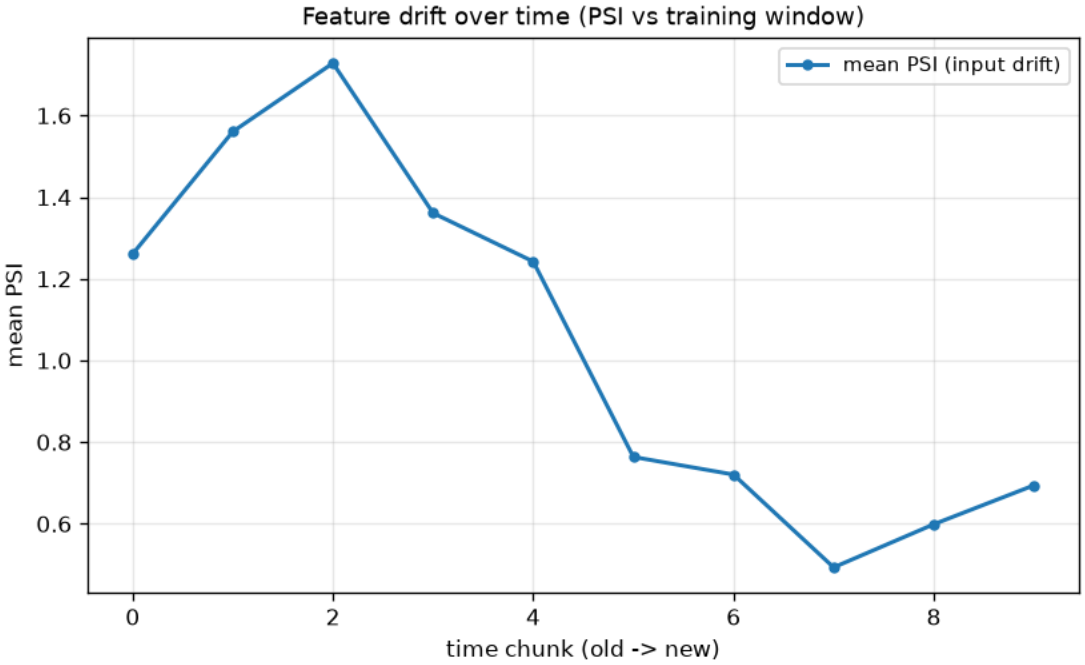
chunk	rows	frozen	adaptive	mean PSI
0	1,813	0.6680	0.6680	1.261
1	1,813	0.6751	0.6746	1.561
2	1,813	0.6299	0.6365	1.728
3	1,813	0.6266	0.6547	1.361
4	1,813	0.7187	0.7253	1.242
5	1,812	0.7230	0.6893	0.764
6	1,812	0.7196	0.7009	0.721
7	1,812	0.8328	0.8306	0.493

chunk	rows	frozen	adaptive	mean PSI
8	1,812	0.8350	0.8262	0.599
9	1,812	0.8179	0.8267	0.694

Chunk 1 is a built-in consistency check: both models are identical there by construction, since neither has seen the chunk yet.



Frozen vs adaptive performance across the held-out period



Input drift (PSI) against the training window

Mean score, frozen model	0.7247
Mean score, adaptive model	0.7233
First third -> last third	- -> -

Performance worsened	no
Worst-drifting feature	-
Peak mean PSI	1.728

Drift verdict

Frozen-model accuracy held up across time (0.668 -> 0.818; drift magnitude -0.171). Peak input drift (mean PSI) = 1.728 (notable). A sliding-retrained model did not beat the frozen one (accuracy 0.723 vs 0.725, gap 0.001 but not statistically significant).

Statistical significance

A score difference is only meaningful if it survives a significance test. Paired predictions on the same held-out rows are compared with the McNemar's test. A p-value below 0.05 means the difference is unlikely to be chance; above it, the two models are statistically indistinguishable - which is itself worth reporting.

comparison	better model	p-value	conclusion
champion vs runner up	logreg_l1	1	not significant
static vs adaptive	static	0.3976	not significant

champion vs runner up: logreg_l1 and logreg_elasticnet disagree on 1 row, split fairly evenly. p=1 >= 0.05, so the difference is NOT statistically significant.

static vs adaptive: static and adaptive disagree on 805 rows, split fairly evenly. p=0.3976 >= 0.05, so the difference is NOT statistically significant.

APPENDIX A

Scope and limitations

Tuning is measured, not assumed

The point of Stage 3 is to quantify the cost/benefit of tuning, not to prove that tuning always wins. On well-conditioned datasets a regularized baseline can match a tuned booster, and this report says so when it happens.

Drift is detected, not explained

PSI and prequential error are cheap, reliable indicators that a distribution has moved. Neither attributes a cause; deeper causal drift analysis is out of scope.

Single-machine scale

The pipeline targets datasets that fit in memory on one machine. It is not a distributed training system.

One split, one seed

Results come from a single temporal split at a fixed random seed. That is the right choice for a time-ordered problem - shuffled k-fold would leak the future - but it means confidence intervals across resamples are not reported, which is why the significance tests above matter.

APPENDIX B

Glossary

ROC-AUC	Probability a random positive is ranked above a random negative. 0.5 is chance, 1.0 is perfect. Insensitive to class imbalance.

RMSE / MAE	Average prediction error in the target's own units. Lower is better; RMSE punishes large misses harder than MAE.
R-squared	Share of the target's variance the model explains. 0 means no better than predicting the mean.
Regularization (L1/L2)	A penalty on coefficient size that trades a little bias for less variance. L1 can zero coefficients outright, performing feature selection; L2 shrinks them smoothly.
Stacking	Training a small meta-model on the base models' predictions so it can learn which to trust where.
Bayesian optimization	Tuning that builds a surrogate model of the search space to choose each next trial, rather than sampling blindly.
SHAP value	A feature's signed contribution to one prediction, derived from cooperative game theory. Contributions sum to the prediction.
PSI	Population Stability Index: how far a feature's distribution has moved from its training reference. Rules of thumb: below 0.1 stable, 0.1-0.25 moderate, above 0.25 material.
Prequential evaluation	Test-then-train: each chunk is predicted before the model is allowed to learn from it, so no future information leaks.
McNemar's test	A paired test for two classifiers on identical rows, based only on the cases where they disagree.

Produced by the TabML-Guard pipeline.

~ Ayush Gole